Untitled67.ipynb ☆ Saving...

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[1] 0s
```
# Seaborn
```

[2]
```
# Visualization library build on matplotlib
```

[3] 0s
```
# import all libraries
```

[4] 1s
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[5]
```
# defult matplotlib
# plt.title
# plt.xlable
# plt.ylable
# plt.xticks(rotation=45)
# plt.savefig()
```

[6] 0s
```
titanic = sns.load_dataset("titanic")
```

[7] 0s
```
titanic.head()
```

| | survived | pclass | sex | age | sibsp | parch | fare | embarked | class | who |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 3 | male | 22.0 | 1 | 0 | 7.2500 | S | Third | mar |
| 1 | 1 | 1 | female | 38.0 | 1 | 0 | 71.2833 | C | First | womar |
| 2 | 1 | 3 | female | 26.0 | 0 | 0 | 7.9250 | S | Third | womar |
| 3 | 1 | 1 | female | 35.0 | 1 | 0 | 53.1000 | S | First | womar |
| 4 | 0 | 3 | male | 35.0 | 0 | 0 | 8.0500 | S | Third | mar |

Next steps: Generate code with titanic | New interactive sheet

[8] 0s
```
titanic["sex"].value_counts()
```

| | count |
|---|---|

| sex | |
|---|---|
| male | 577 |
| female | 314 |

dtype: int64

[9] ✓ 0s

```
titanic["sex"].value_counts().reset_index()
```

| | sex | count |
|---|---|---|
| 0 | male | 577 |
| 1 | female | 314 |

[10] ✓ 0s

```
# Count Plot
```

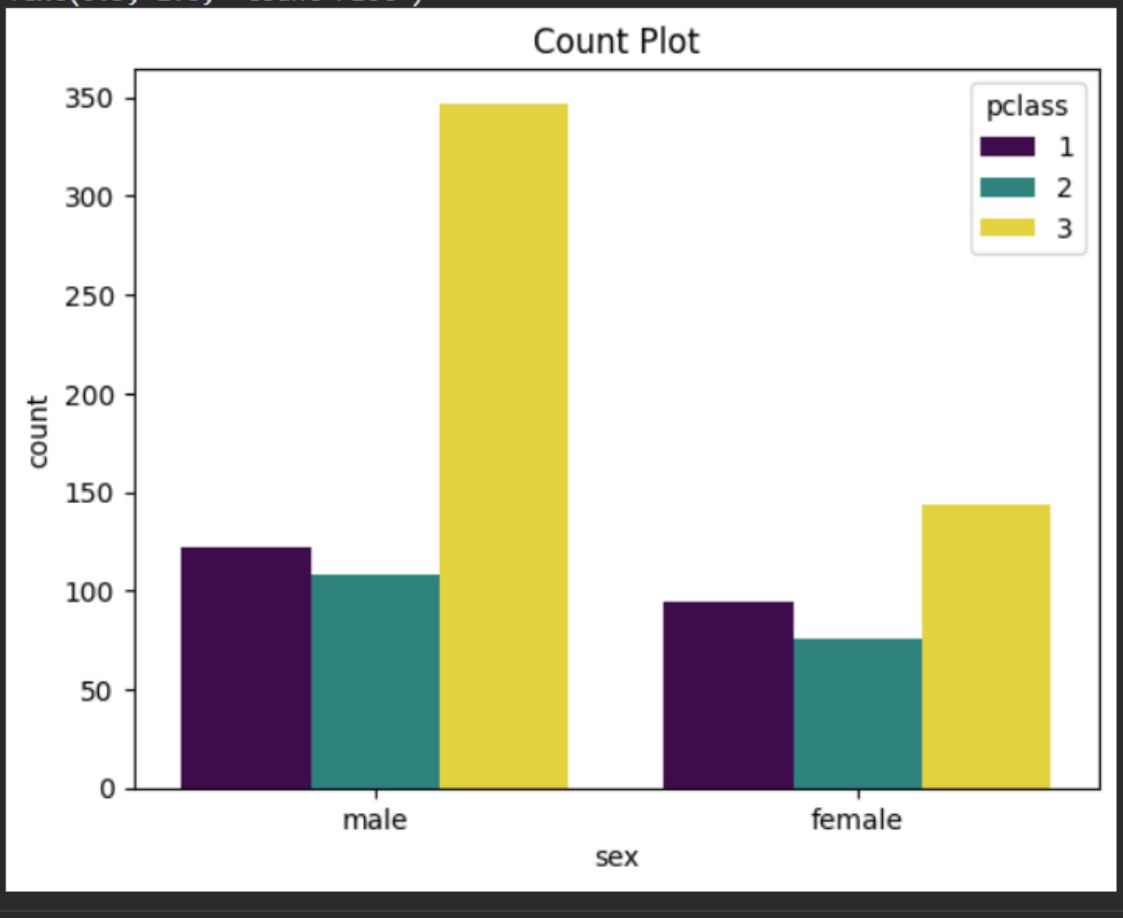
[11] ✓ 0s

```
# syntax -
# sns.plotname(data=titanic, x="columnname",
# y="columnname" , hue="column name",
# palette ="color name")
```

[17] ✓ 0s

```
sns.countplot(data= titanic,
              x="sex" , hue= "pclass" , palette="viridis")
plt.title("Count Plot")
```

```
Text(0.5, 1.0, 'Count Plot')
```



[18] ✓ 0s

```
# Bar Char
```

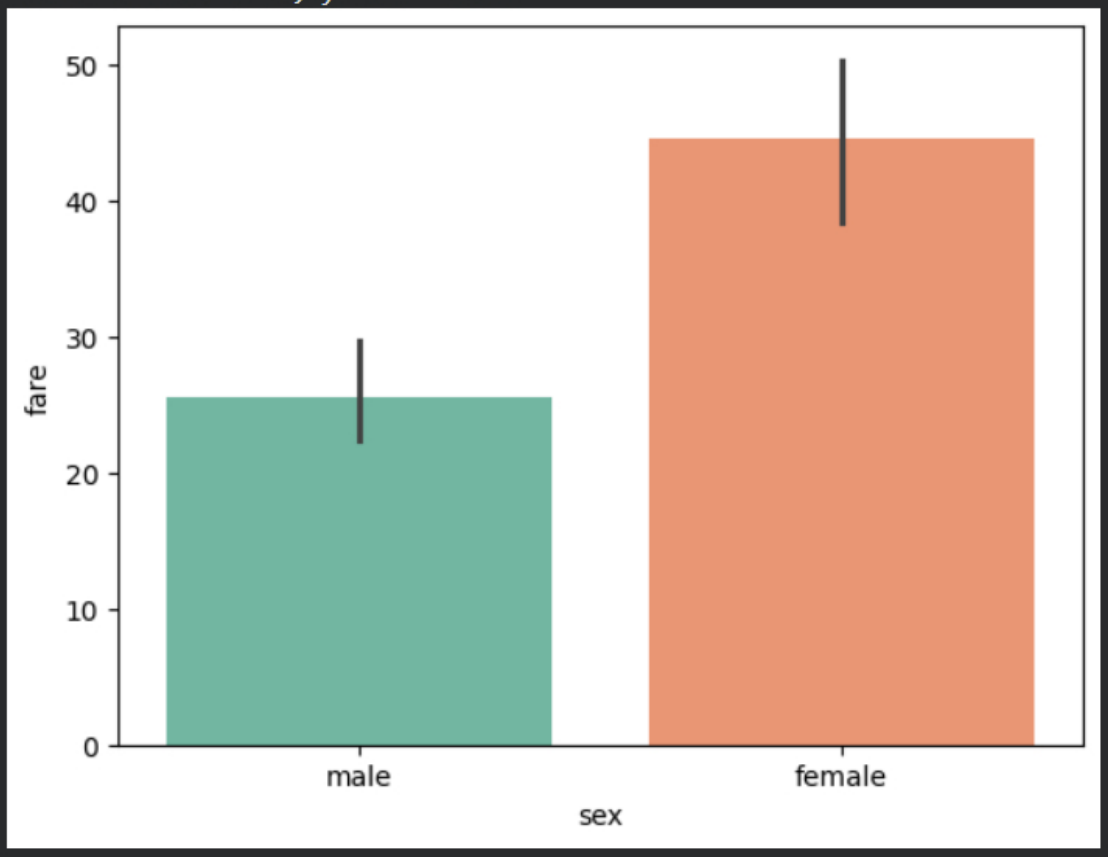
[23] ✓ 0s

```
sns.barplot(data= titanic, x= "sex", y="fare", palette="Set2")
```

```
/tmp/ipykernel_2656/40689280.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in

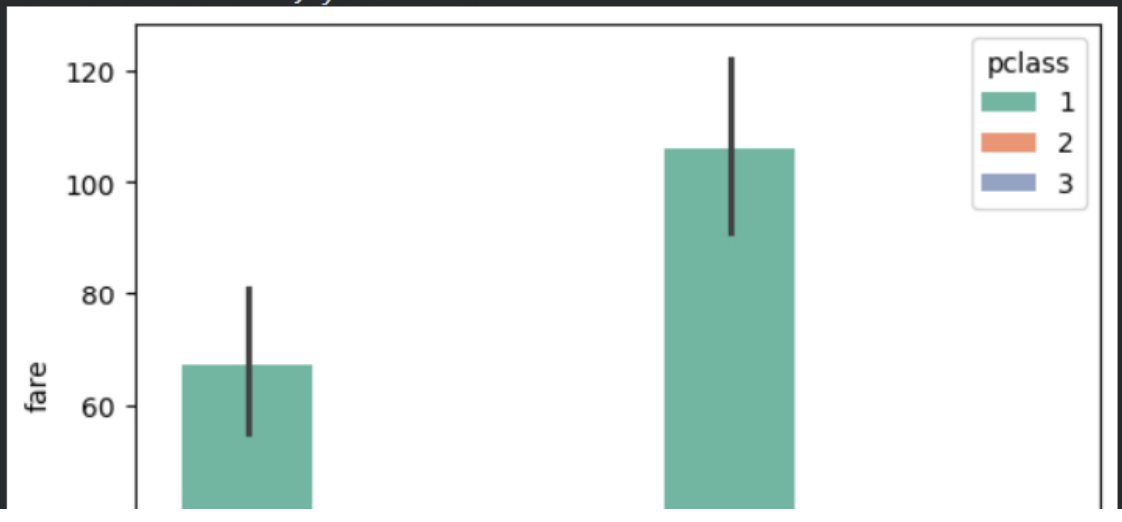
  sns.barplot(data= titanic, x= "sex", y="fare", palette="Set2")
<Axes: xlabel='sex', ylabel='fare'>
```

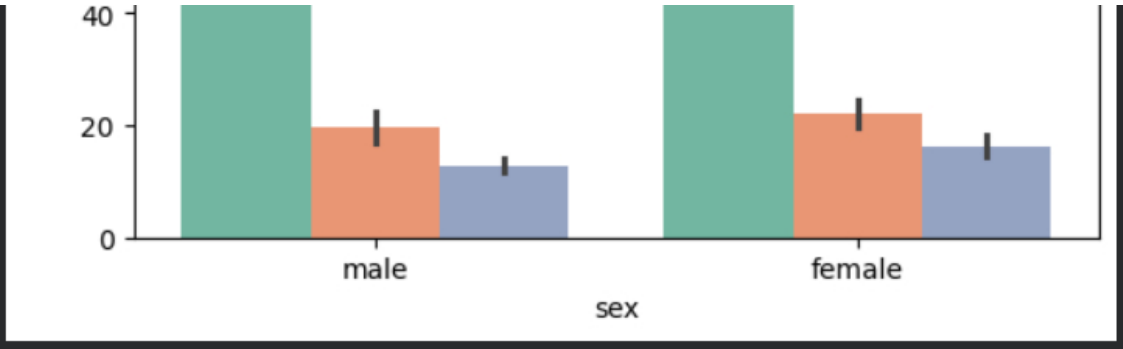



[24] ✓ 0s

```
sns.barplot(data= titanic, x= "sex", y="fare",
            hue = "pclass", palette="Set2")
```

```
<Axes: xlabel='sex', ylabel='fare'>
```

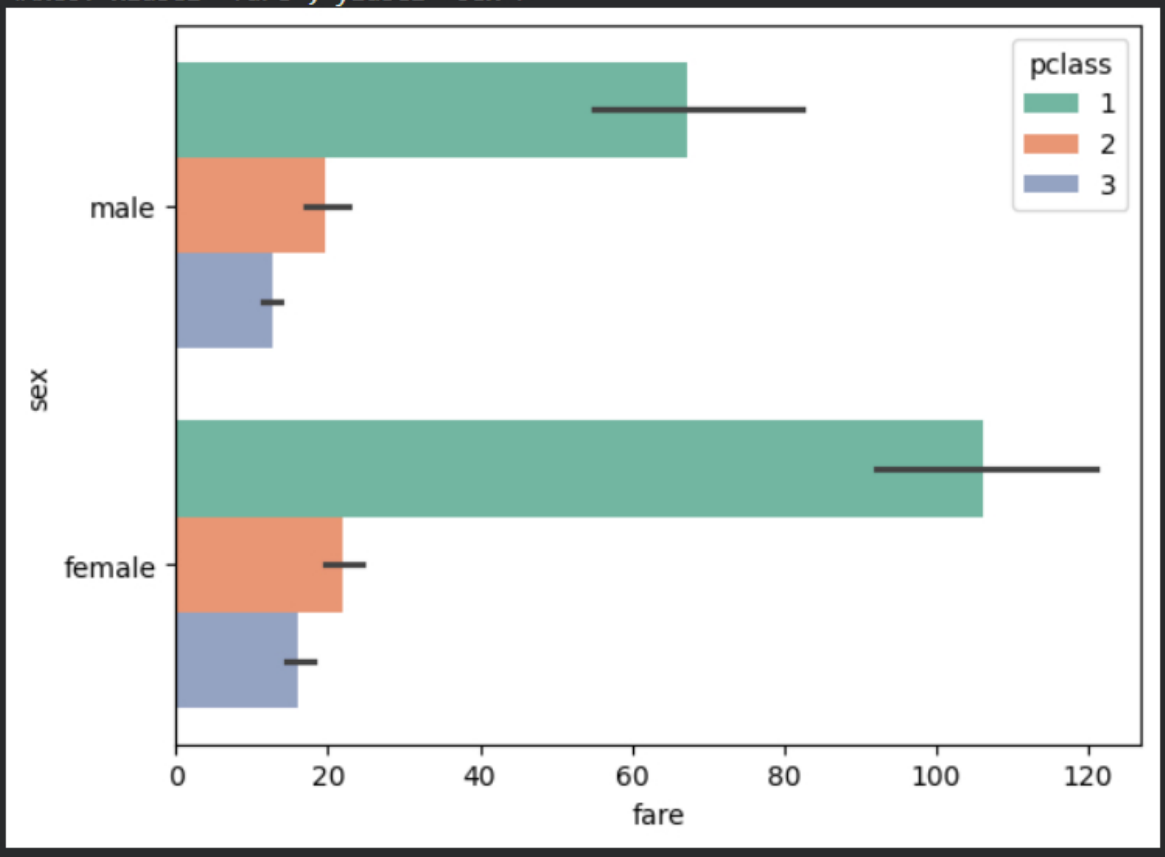





[25]
✓ 0s

```
sns.barplot(data= titanic, x= "fare", y="sex",
            hue = "pclass", palette="Set2")
```

```
<Axes: xlabel='fare', ylabel='sex'>
```



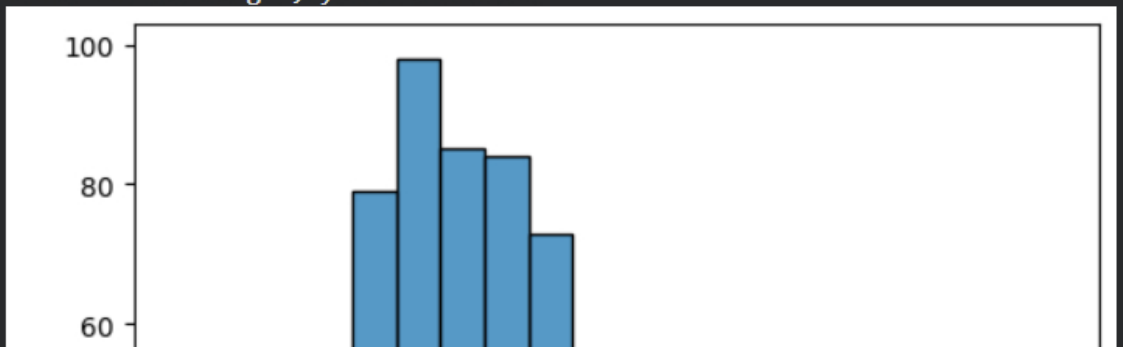

[26]
✓ 0s

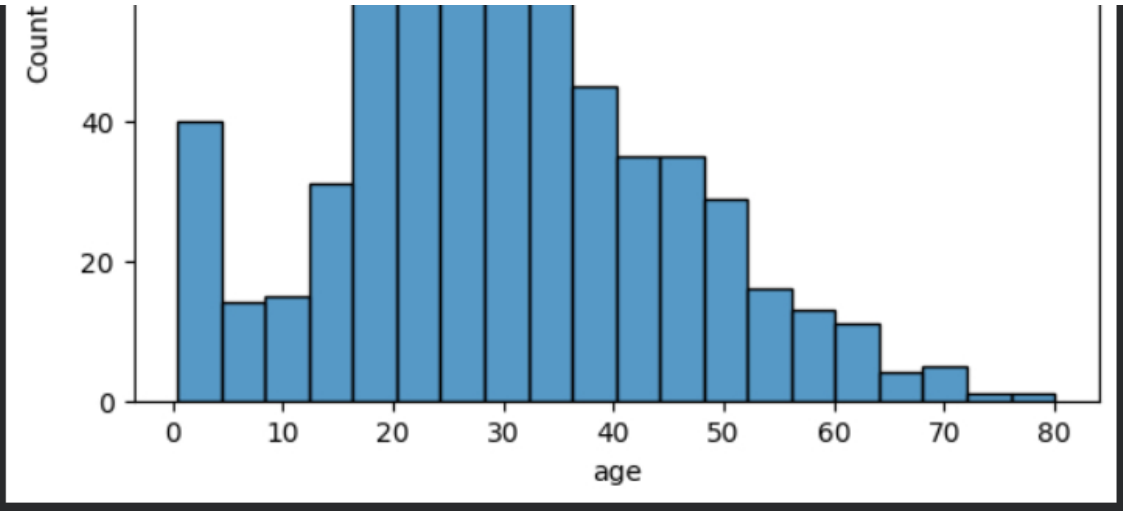
```
# histogram
```

[29]
✓ 0s

```
sns.histplot(data= titanic, x ="age" )
```

```
<Axes: xlabel='age', ylabel='Count'>
```

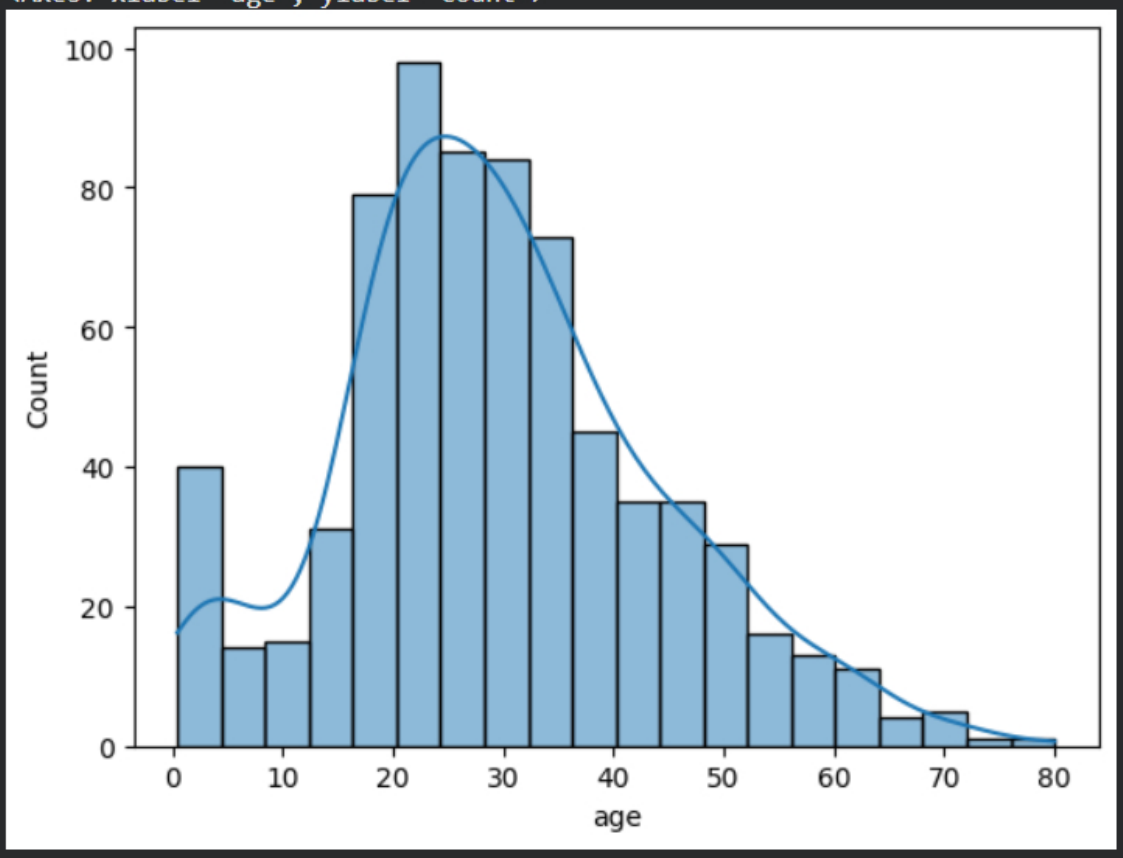





[30]
✓ 0s

```
sns.histplot(data= titanic, x ="age" , kde= True)
```

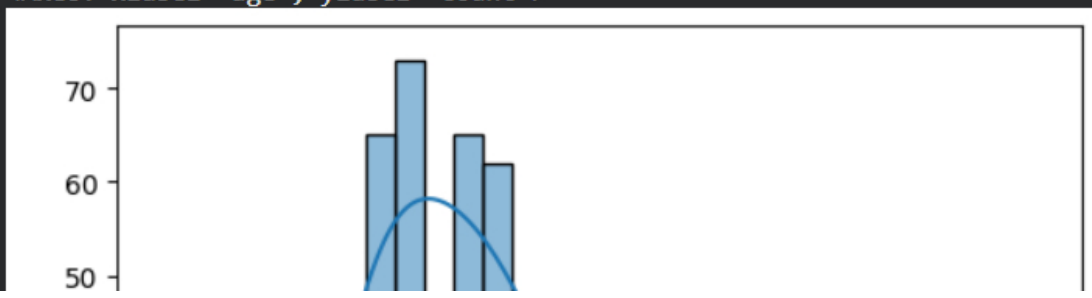
```
<Axes: xlabel='age', ylabel='Count'>
```

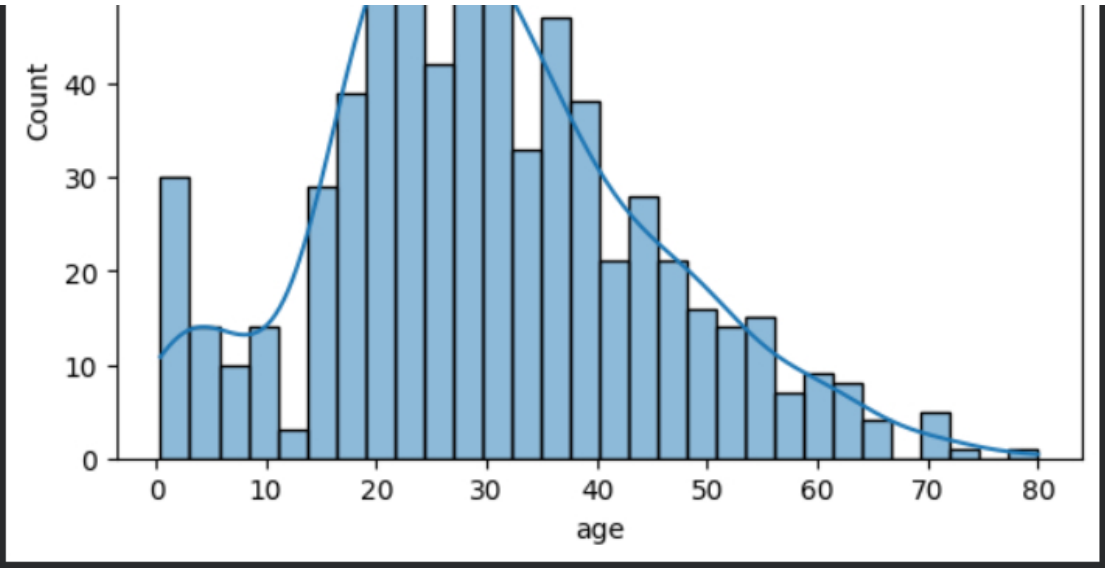



[31]
✓ 0s

```
sns.histplot(data= titanic, x ="age" , kde= True , bins =30)
```

```
<Axes: xlabel='age', ylabel='Count'>
```

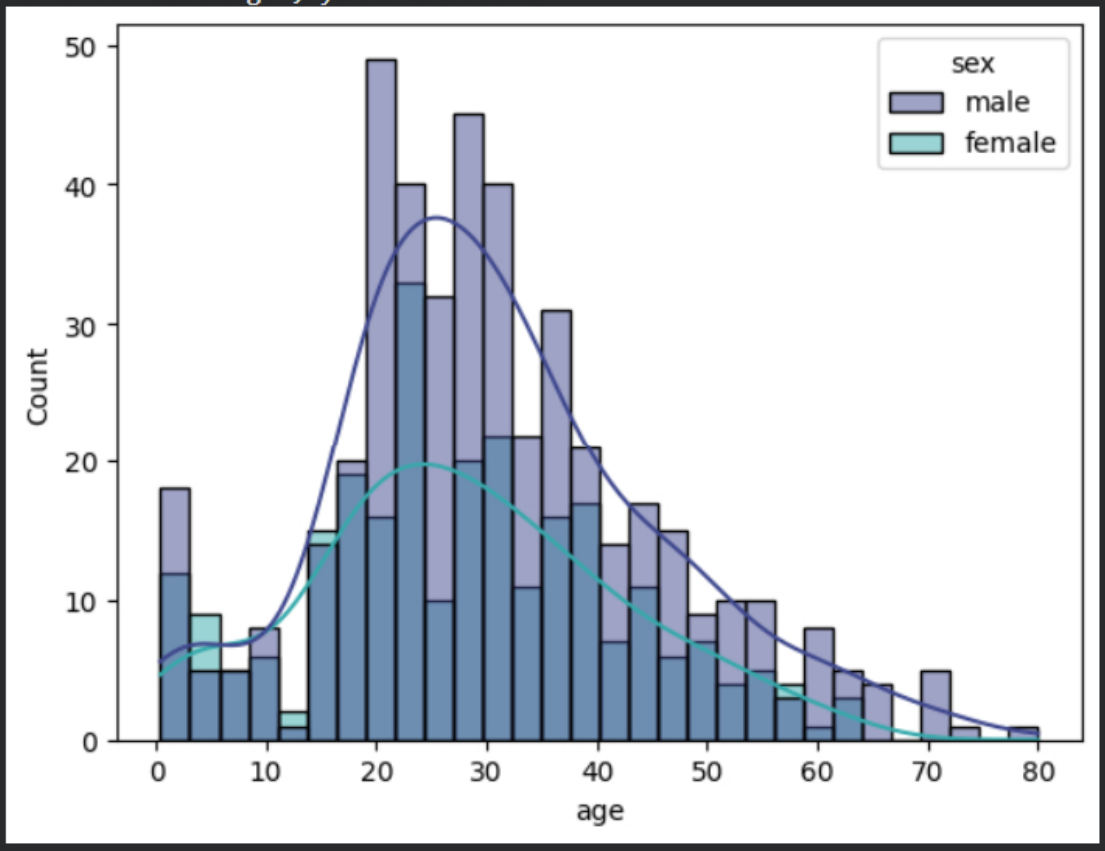





[33]
✓ 0s

```
sns.histplot(data= titanic, x ="age" ,
             hue = "sex", kde= True , bins =30 , palette="mako")
```

```
<Axes: xlabel='age', ylabel='Count'>
```




[34]
✓ 0s

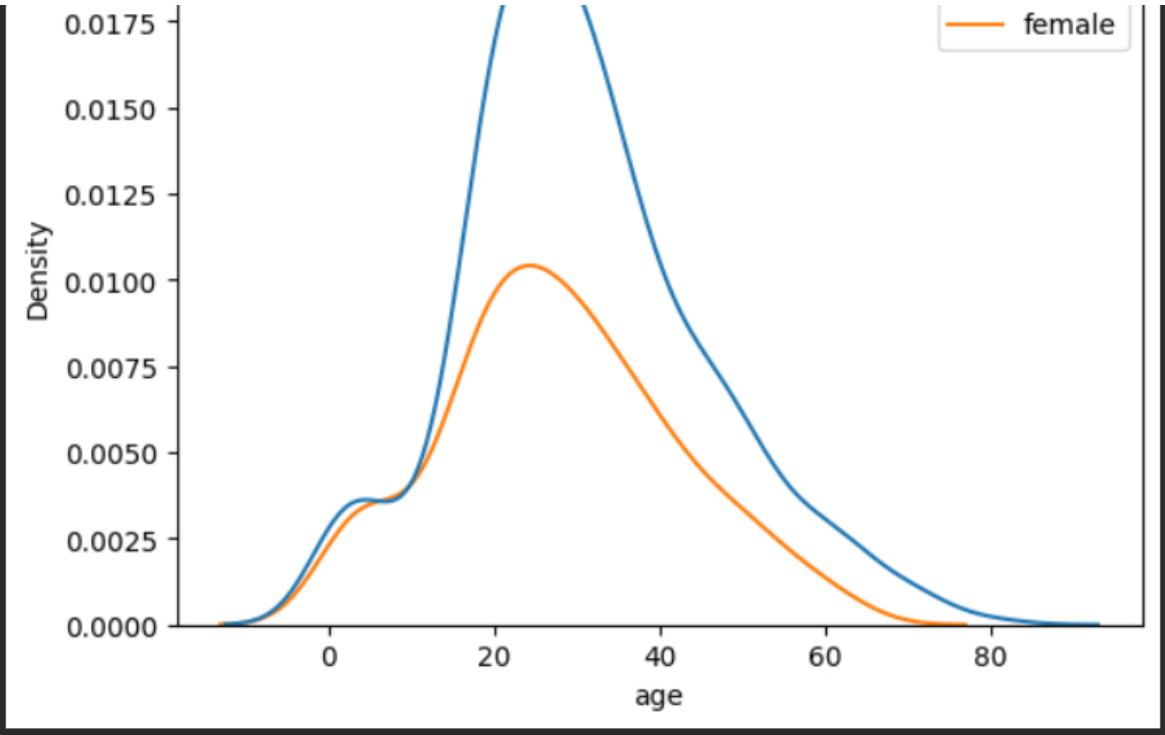
```
# kde plot
```

[35]
✓ 0s

```
sns.kdeplot(data= titanic, x="age", hue="sex")
```

```
<Axes: xlabel='age', ylabel='Density'>
```





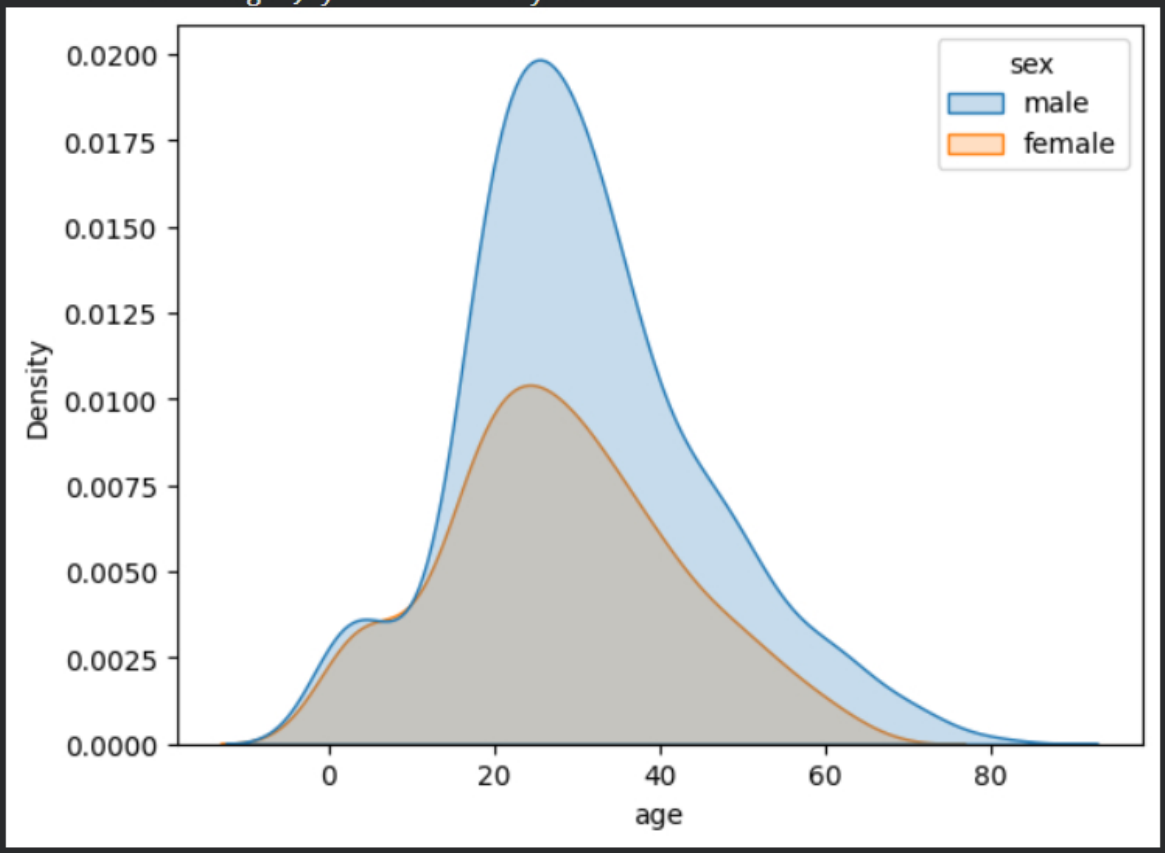

[37]
✓ 0s

```
# area plot
```

[38]
✓ 0s

```
sns.kdeplot(data= titanic, x="age", hue="sex", fill =True)
```

```
<Axes: xlabel='age', ylabel='Density'>
```



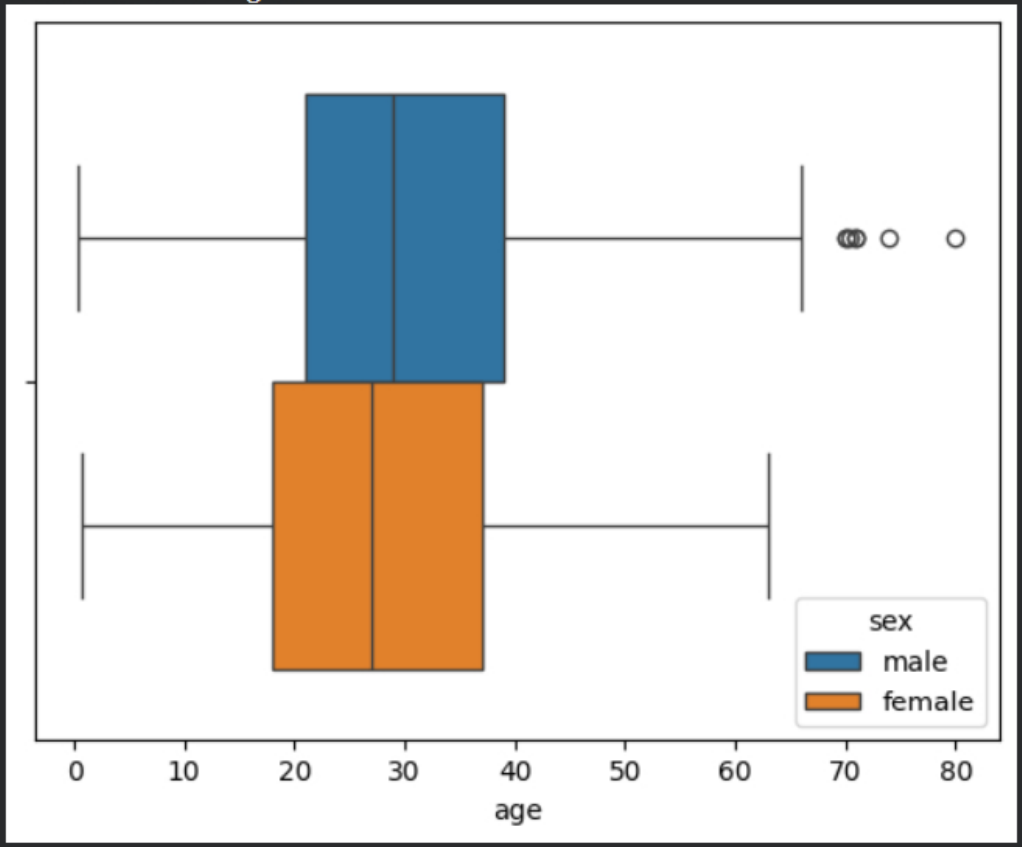

[39]
✓ 0s

```
# BOXPLOT
```

[40]
✓ 0s

```
sns.boxplot(data= titanic, x="age" , hue="sex")
```

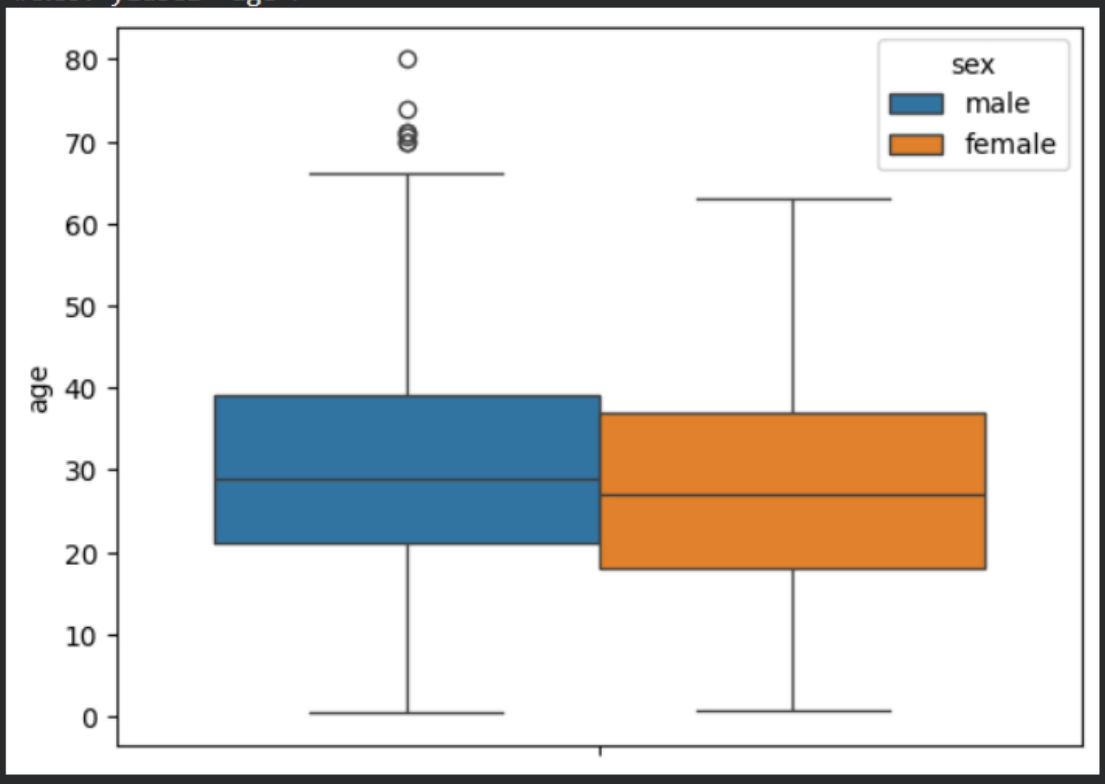
```
<Axes: xlabel='age'>
```




[41]
✓ 0s

```
sns.boxplot(data= titanic, y="age" , hue="sex")
```

```
<Axes: ylabel='age'>
```



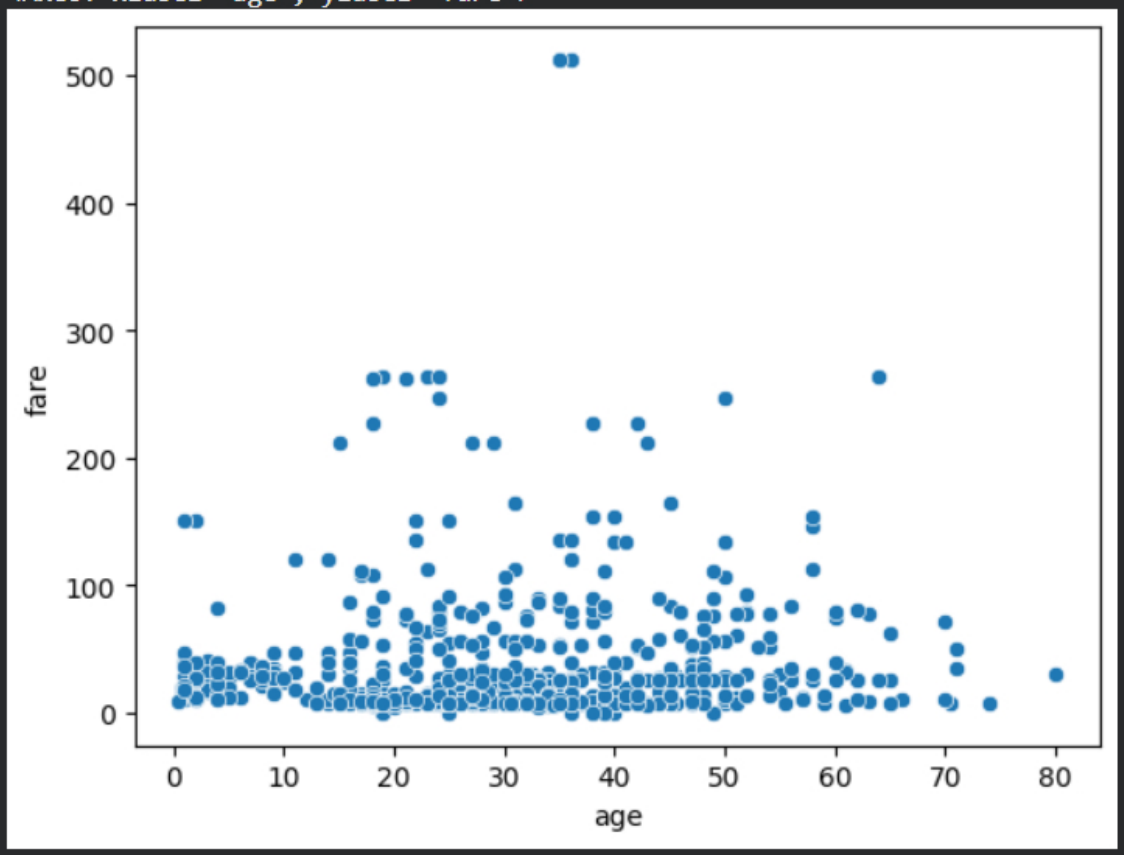
[42] ✓ 0s

```
# scatter plot
```

[43] ✓ 0s

```
sns.scatterplot(data= titanic , x="age", y="fare")
```

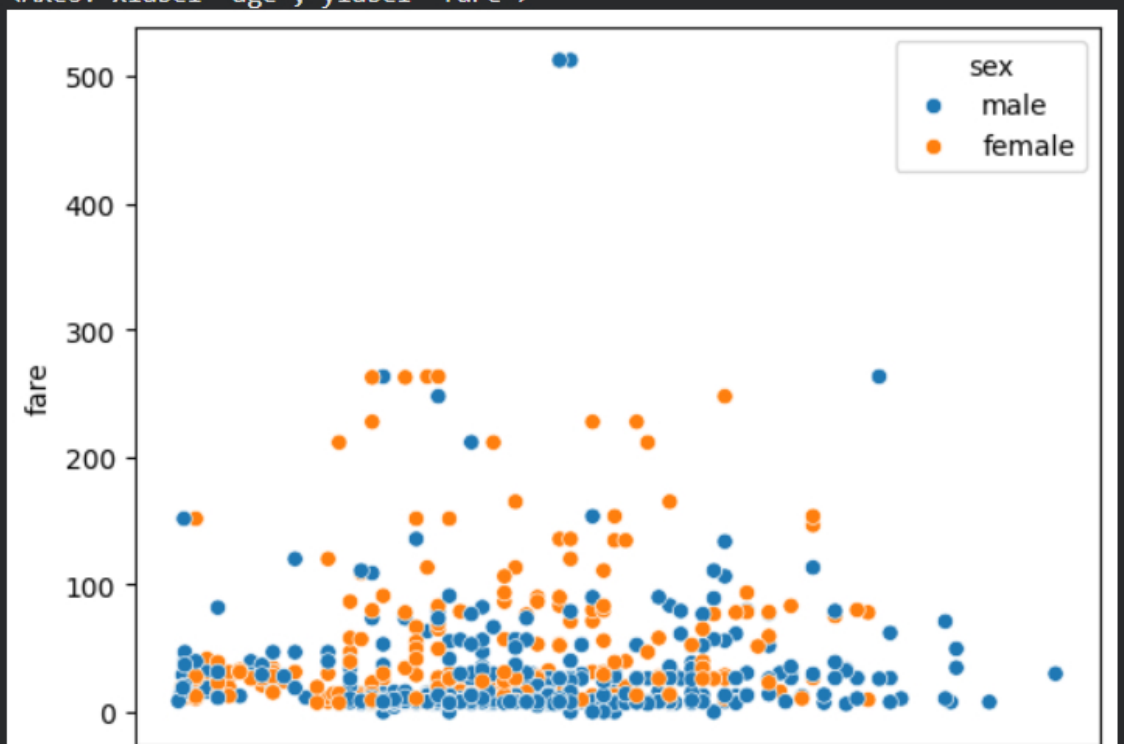
```
<Axes: xlabel='age', ylabel='fare'>
```

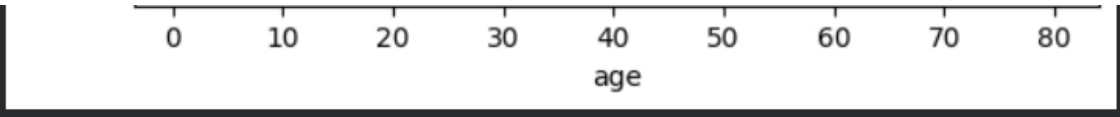



[44] ✓ 0s

```
sns.scatterplot(data= titanic , x="age", y="fare",
                hue="sex")
```

```
<Axes: xlabel='age', ylabel='fare'>
```

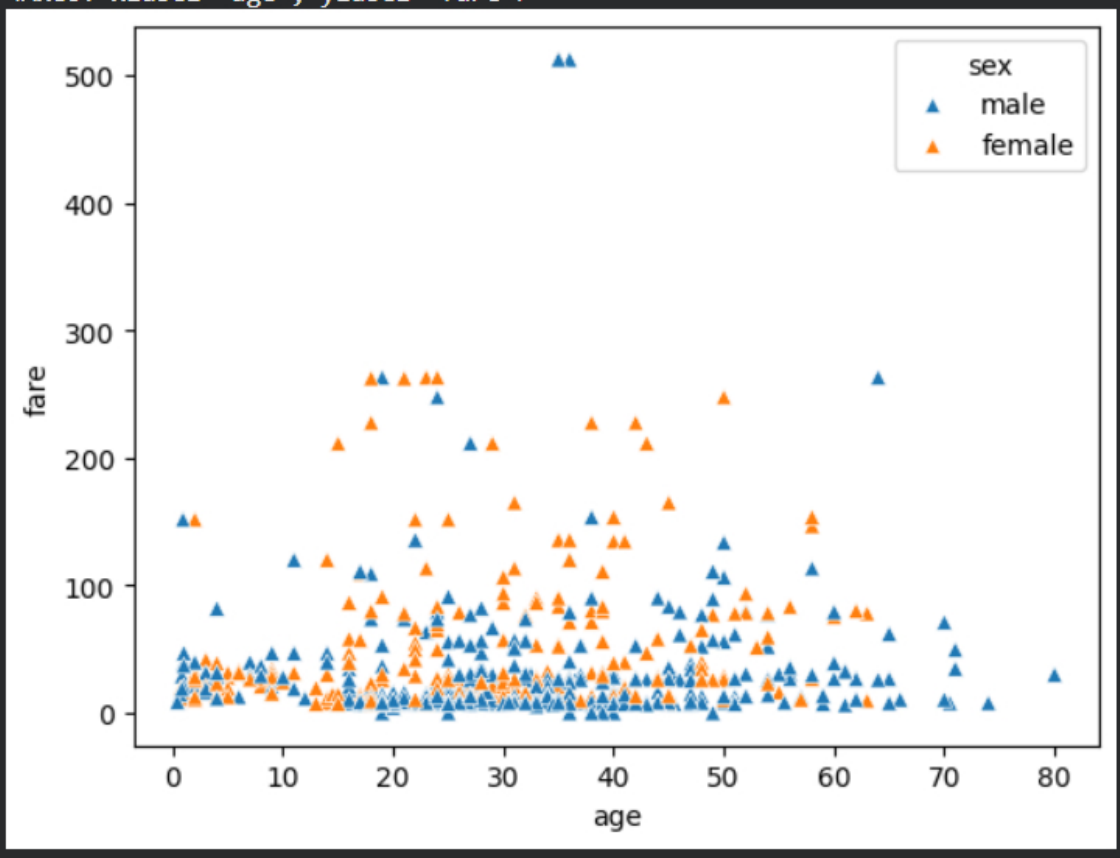





[45]
✓ 0s

```
sns.scatterplot(data= titanic , x="age", y="fare",
                hue="sex", marker ="^")
```

```
<Axes: xlabel='age', ylabel='fare'>
```



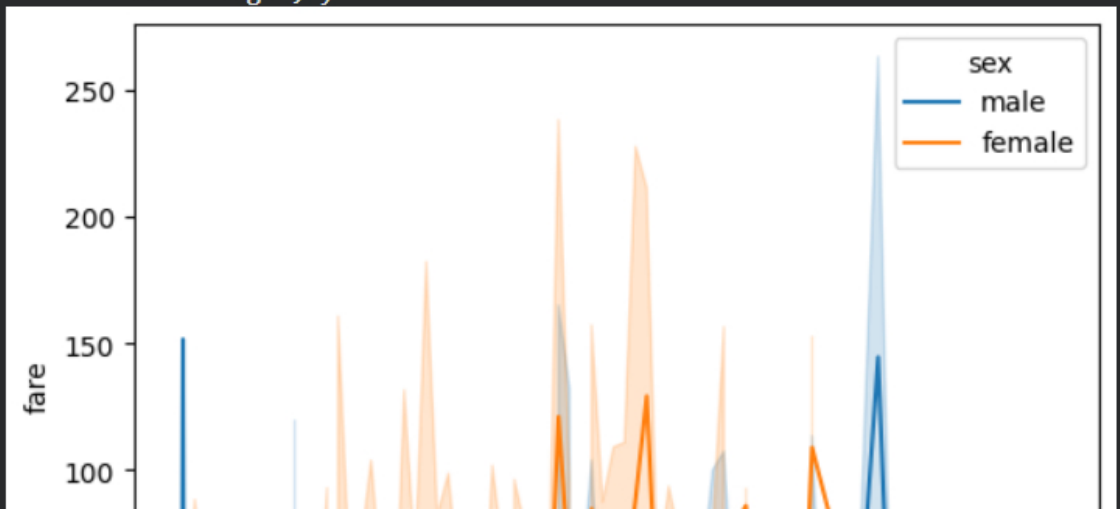

[46]
✓ 0s

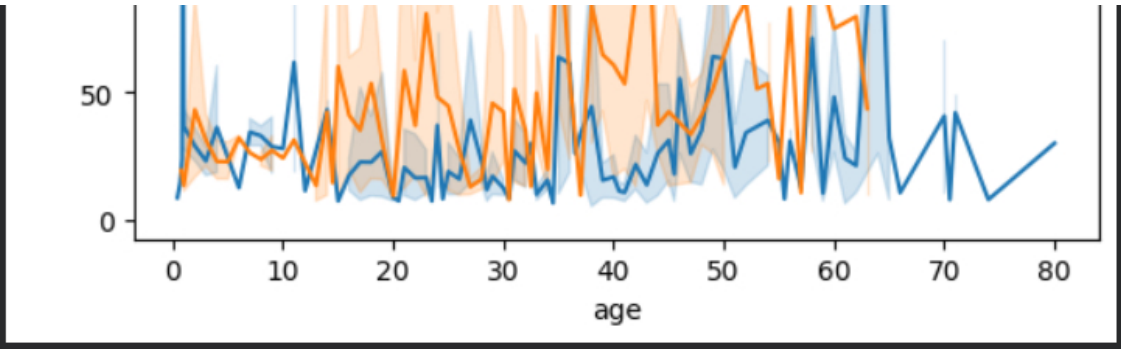
```
# lineplot
```

[47]
✓ 3s

```
sns.lineplot(data= titanic, x="age", y="fare", hue="sex")
```

```
<Axes: xlabel='age', ylabel='fare'>
```






[49] ✓ 0s

```
# correlation - if one variable increases dose
# the other variable increases or not ?
# correlation matrix
```

[50] ✓ 0s

```
corr = titanic.select_dtypes("number").corr()
```

[51] ✓ 0s

```
corr
```

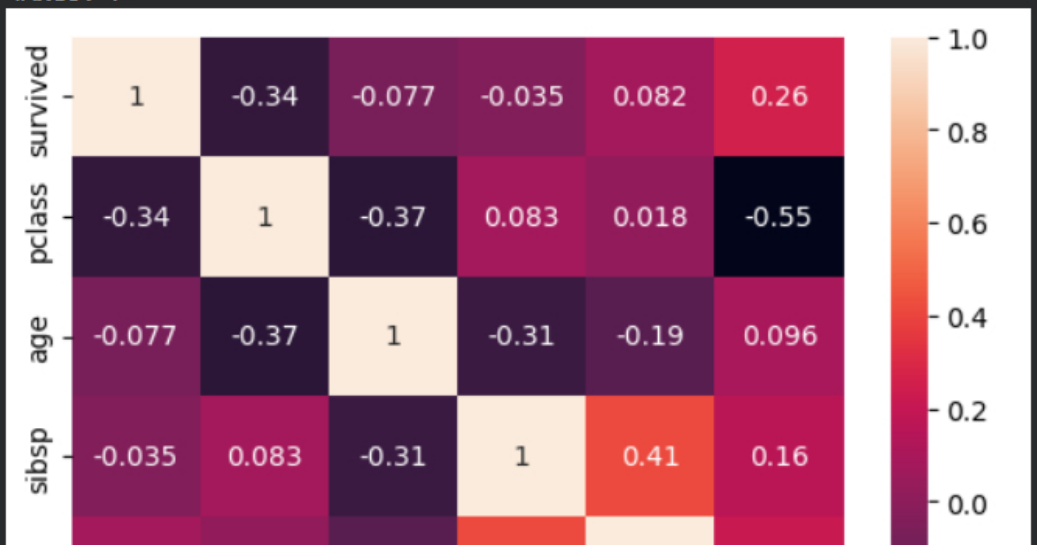
| | survived | pclass | age | sibsp | parch | fare |
|---|---|---|---|---|---|---|
| **survived** | 1.000000 | -0.338481 | -0.077221 | -0.035322 | 0.081629 | 0.257307 |
| **pclass** | -0.338481 | 1.000000 | -0.369226 | 0.083081 | 0.018443 | -0.549500 |
| **age** | -0.077221 | -0.369226 | 1.000000 | -0.308247 | -0.189119 | 0.096067 |
| **sibsp** | -0.035322 | 0.083081 | -0.308247 | 1.000000 | 0.414838 | 0.159651 |
| **parch** | 0.081629 | 0.018443 | -0.189119 | 0.414838 | 1.000000 | 0.216225 |
| **fare** | 0.257307 | -0.549500 | 0.096067 | 0.159651 | 0.216225 | 1.000000 |

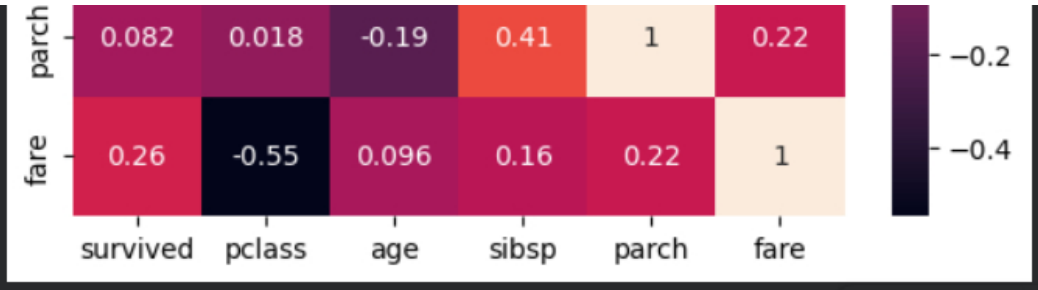
Next steps: Generate code with corr | New interactive sheet

[53] ✓ 0s

```
sns.heatmap(corr, annot=True)
```

```
<Axes: >
```






[ ]

Start coding or generate with AI.

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Variables Terminal 11:11 AM Python 3